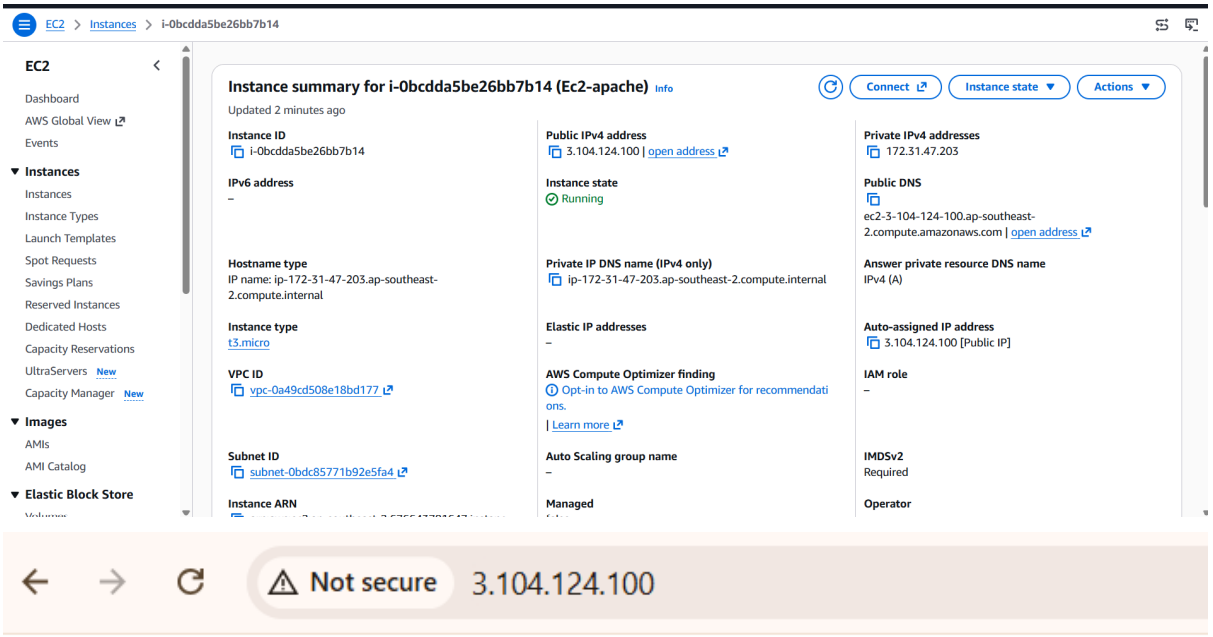
1. **Launch one EC2 using Amazon Linux 2 image and add a script in user data to install Apache**

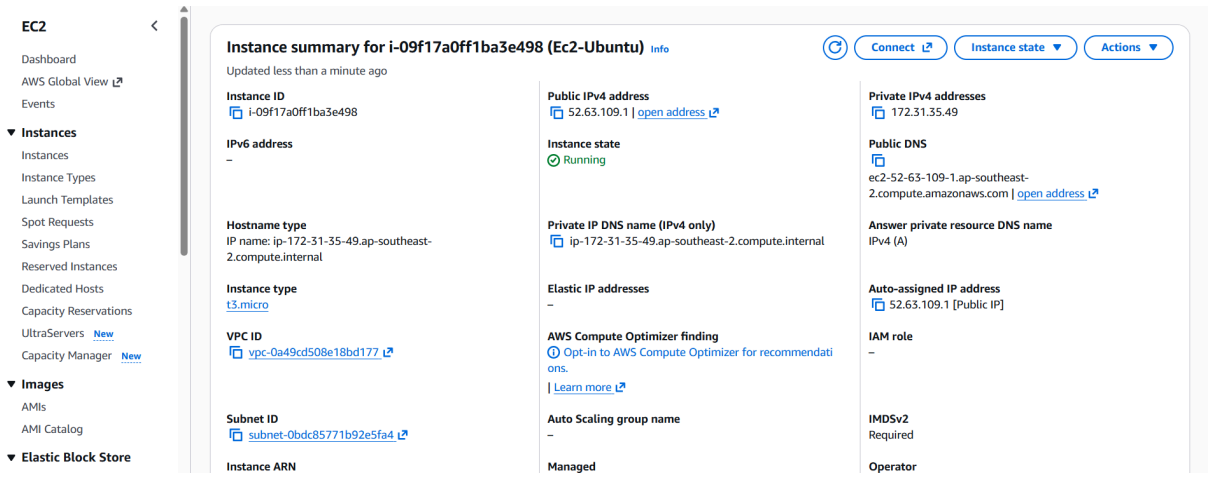


Apache installed successfully using User Data

Steps I Used

1. I opened the AWS Management Console and navigated to EC2 → Launch Instance.
2. I selected the Amazon Linux 2 AMI.
3. I selected the required instance type and key pair.
4. I configured the Security Group and allowed SSH (22) and HTTP (80).
5. I opened Advanced details and went to the User Data section.
6. I added a script to update packages and install Apache (httpd).
7. I added commands in the script to start and enable the Apache service.
8. I launched the EC2 instance.
9. After the instance was running, I copied its Public IPv4 address.
10. I opened the public IP in a browser and verified that the Apache web server was running.

2. **Launch one EC2 using Ubuntu image and add a script in user data to install Nginx.**



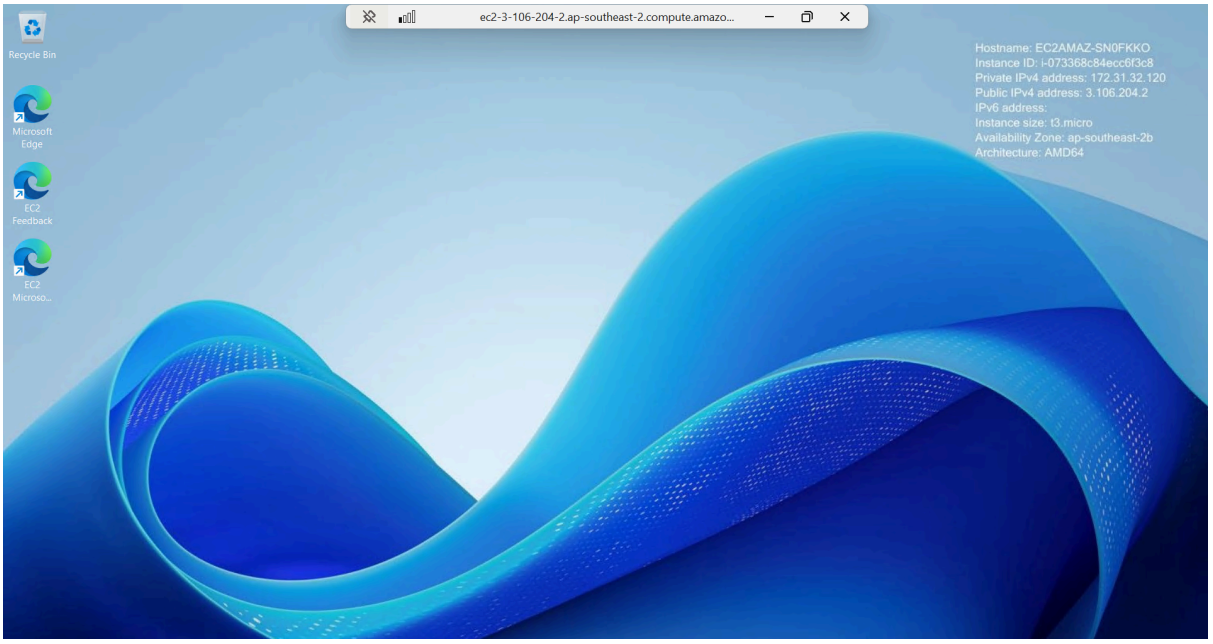

```
ubuntu@ip-172-31-35-49:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-10-28 10:22:49 UTC; 54s ago
 Invocation: 155f85c0551d44f7a4905d50c44a1bfe
       Docs: man:nginx(8)
   Main PID: 1472 (nginx)
      Tasks: 3 (limit: 627)
     Memory: 3M (peak: 7M)
        CPU: 43ms
     CGroup: /system.slice/nginx.service
             ├─1472 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             ├─1475 "nginx: worker process"
             └─1476 "nginx: worker process"

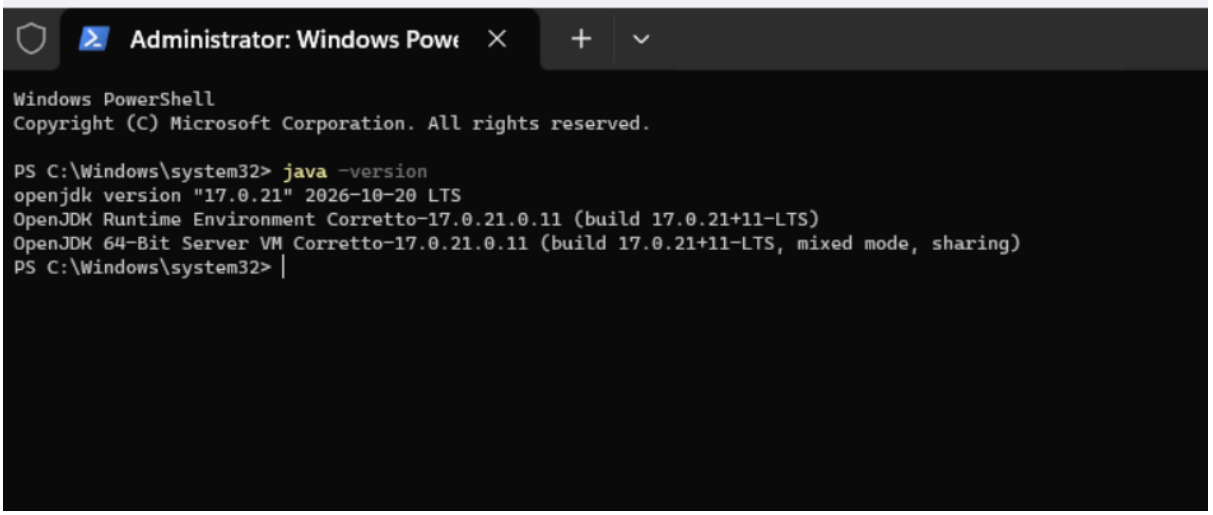
Oct 28 10:22:49 ip-172-31-35-49 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Oct 28 10:22:49 ip-172-31-35-49 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
ubuntu@ip-172-31-35-49:~$ curl http://localhost
Nginx installed successfully using User Data
ubuntu@ip-172-31-35-49:~$
```

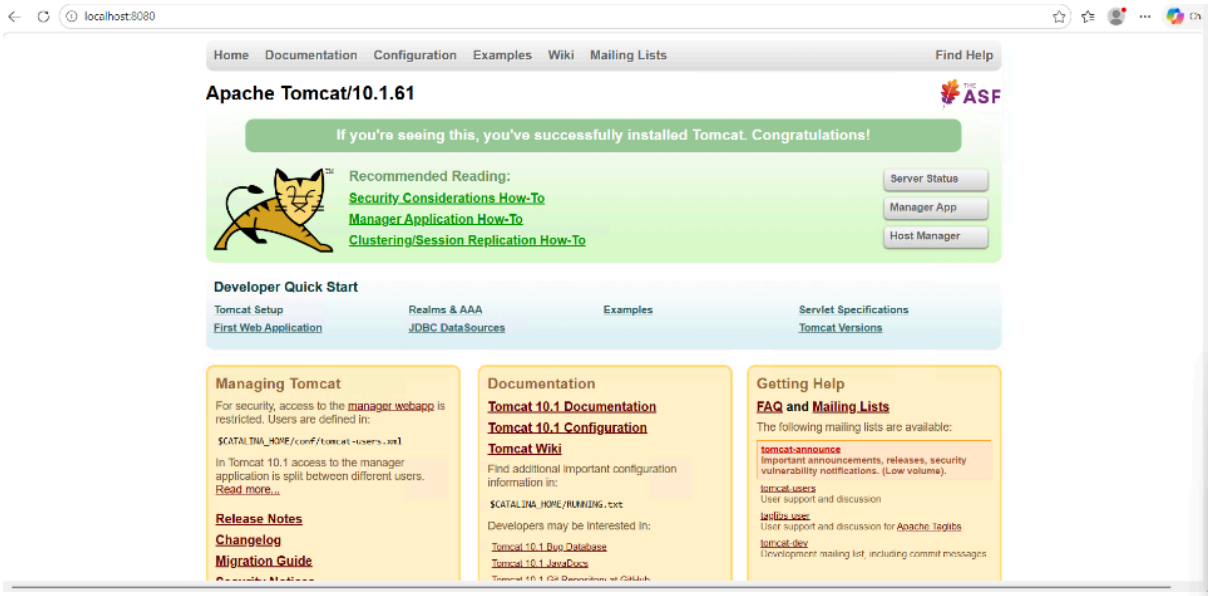
Steps I Used

1. I opened AWS Console → EC2 → Launch Instance.
2. I selected the Ubuntu AMI.
3. I selected the required instance type and key pair.
4. I configured the Security Group and allowed SSH (22) and HTTP (80).
5. I opened Advanced details → User Data.
6. I added a script to update packages and install Nginx.
7. I added commands to start and enable the Nginx service.
8. I launched the EC2 instance.
9. I copied the instance's Public IPv4 address.
10. I opened the public IP in a browser and verified that Nginx was running.

3. Launch one Windows server and install Tomcat on Windows.


ec2-3-106-204-2.ap-southeast-2.compute.amazo...
Recycle Bin
Microsoft Edge
EC2 Feedback
EC2 Microso...
Hostname: EC2AMAZ-SN0FKKO
Instance ID: i-073368c84ecc6f3c8
Private IPv4 address: 172.31.32.120
Public IPv4 address: 3.106.204.2
IPv6 address:
Instance size: t3.micro
Availability Zone: ap-southeast-2b
Architecture: AMD64

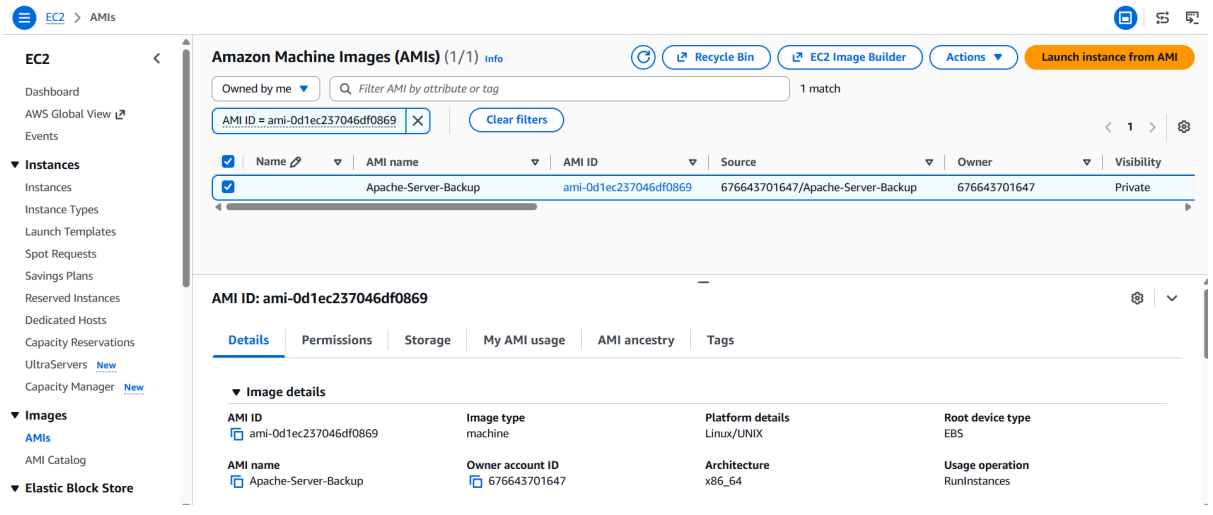

Administrator: Windows Powe
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.
PS C:\Windows\system32> java -version
openjdk version "17.0.21" 2026-10-20 LTS
OpenJDK Runtime Environment Corretto-17.0.21.0.11 (build 17.0.21+11-LTS)
OpenJDK 64-Bit Server VM Corretto-17.0.21.0.11 (build 17.0.21+11-LTS, mixed mode, sharing)
PS C:\Windows\system32>

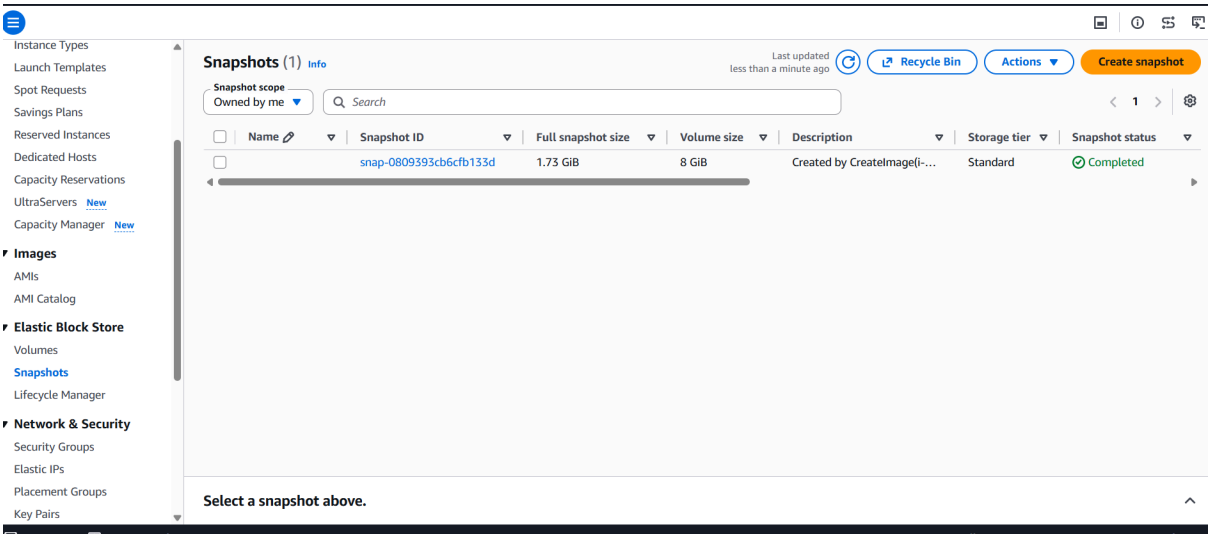

localhost:8080
Home
Documentation
Configuration
Examples
Wiki
Mailing Lists
Find Help
Apache Tomcat/10.1.61
If you're seeing this, you've successfully installed Tomcat. Congratulations!
Recommended Reading:
Security Considerations How-To
Manager Application How-To
Clustering/Session Replication How-To
Server Status
Manager App
Host Manager
Developer Quick Start
Tomcat Setup
First Web Application
Realms & AAA
JDBC DataSources
Examples
Servlet Specifications
Tomcat Versions
Managing Tomcat
Documentation
Tomcat 10.1 Documentation
Tomcat 10.1 Configuration
Tomcat Wiki
Getting Help
FAQ and Mailing Lists
Release Notes
Changelog
Migration Guide

Steps I Used

1. I connected to the Windows EC2 server using RDP.
2. I opened PowerShell inside the Windows EC2 server.
3. I verified that Java was installed.
4. I navigated to the Tomcat bin directory.
5. I started the Tomcat server.
6. I verified that Tomcat was listening on port 8080.
7. I checked Tomcat from the Windows EC2 server.
8. I opened the EC2 Public IPv4 address with port 8080 in my local browser.
9. I verified that the Apache Tomcat welcome page was displayed.

4.Take a snapshot of the instance created in Task 1



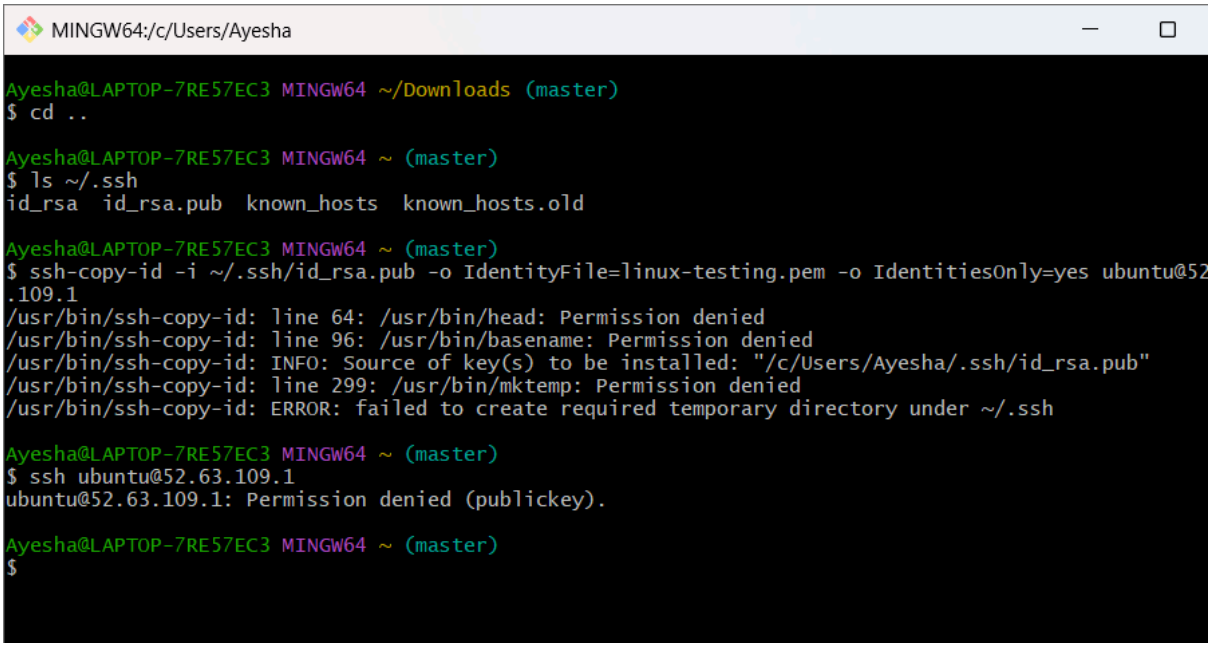


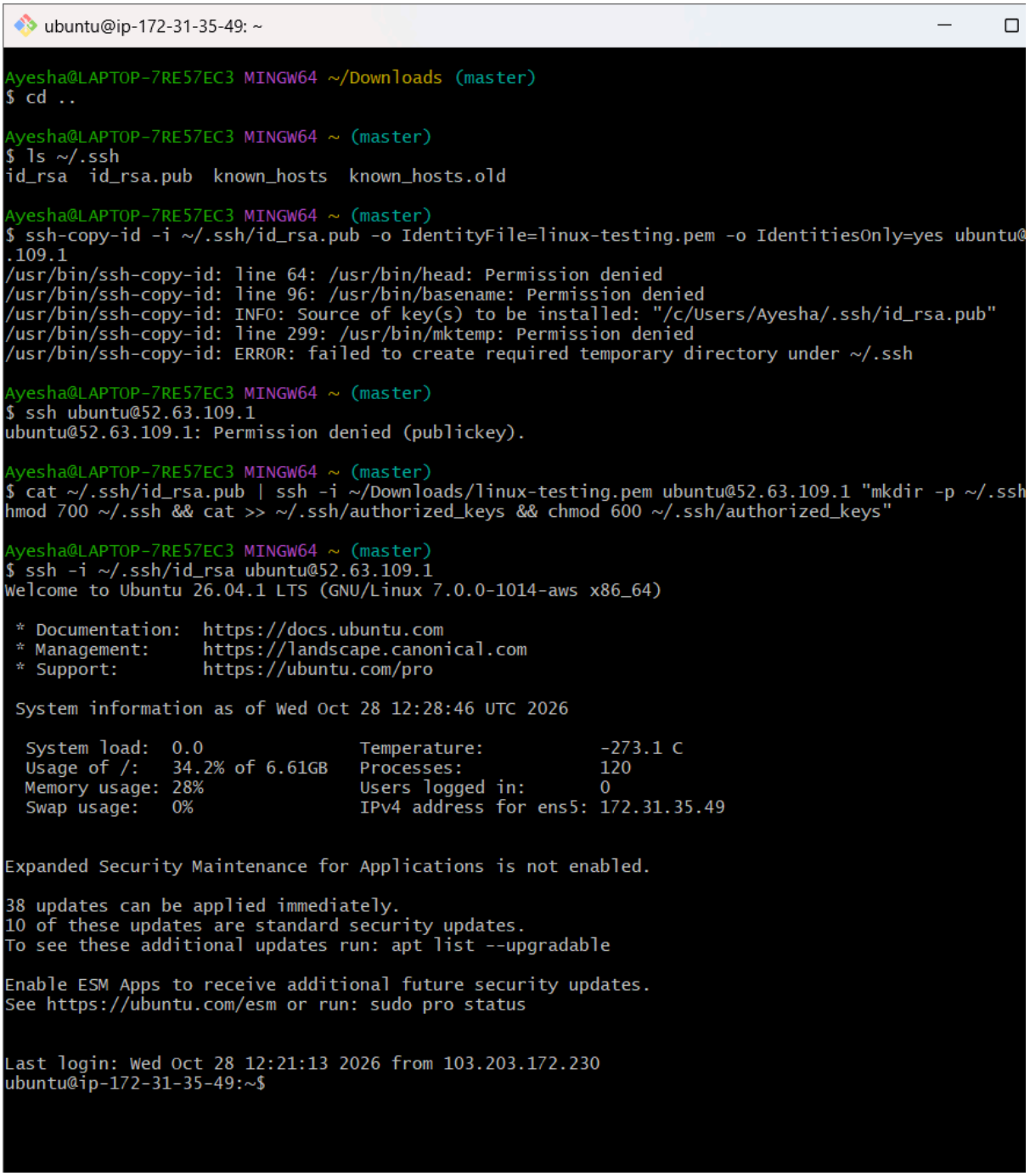
Steps to Take a Snapshot

1. Open AWS Console → EC2 → Instances.
2. Select the Amazon Linux 2 EC2 instance that you created with the Apache User Data script.
3. Make sure the instance is Running and the status checks are complete.
4. Click Actions → Image and templates → Create image.
5. Enter an image name, for example: Apache-Server-Backup
6. Add a description if required.
7. Keep the default volume settings.
8. Click Create image.
9. Go to EC2 → AMIs and verify that the AMI is being created.
10. Go to EC2 → Snapshots and verify the snapshot created from the instance's EBS volume.
11. Wait until the snapshot status becomes Completed.

Important: In AWS, creating an AMI (Create image) automatically creates snapshots of the instance's EBS volumes. If your task specifically says "Take a snapshot", you can also select the instance's EBS volume → Actions → Create snapshot.

5. Assign passwordless authentication for the EC2 created in Task 2.

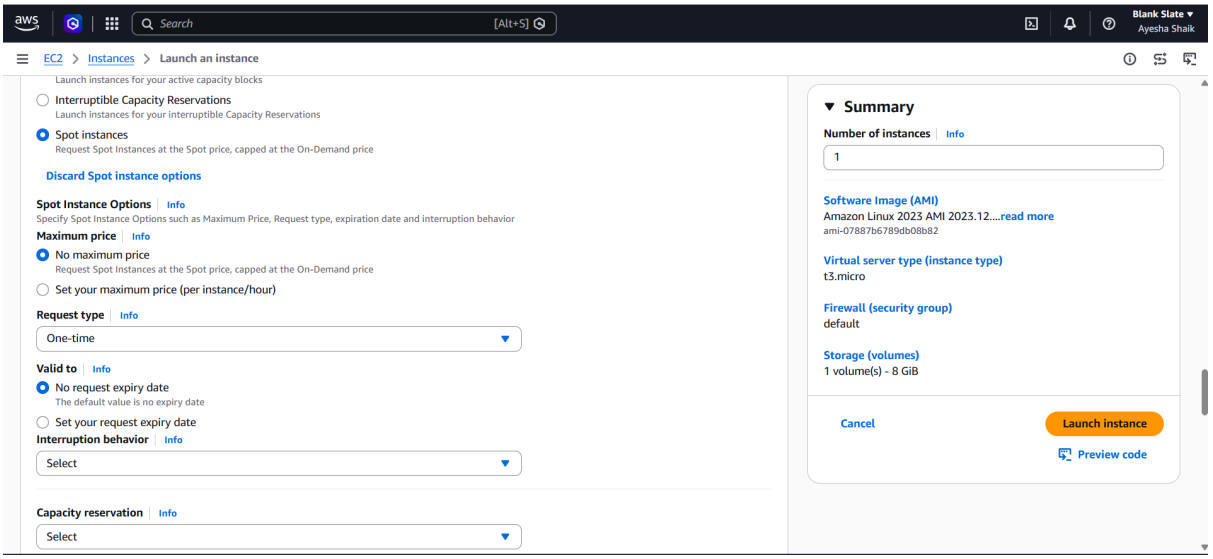


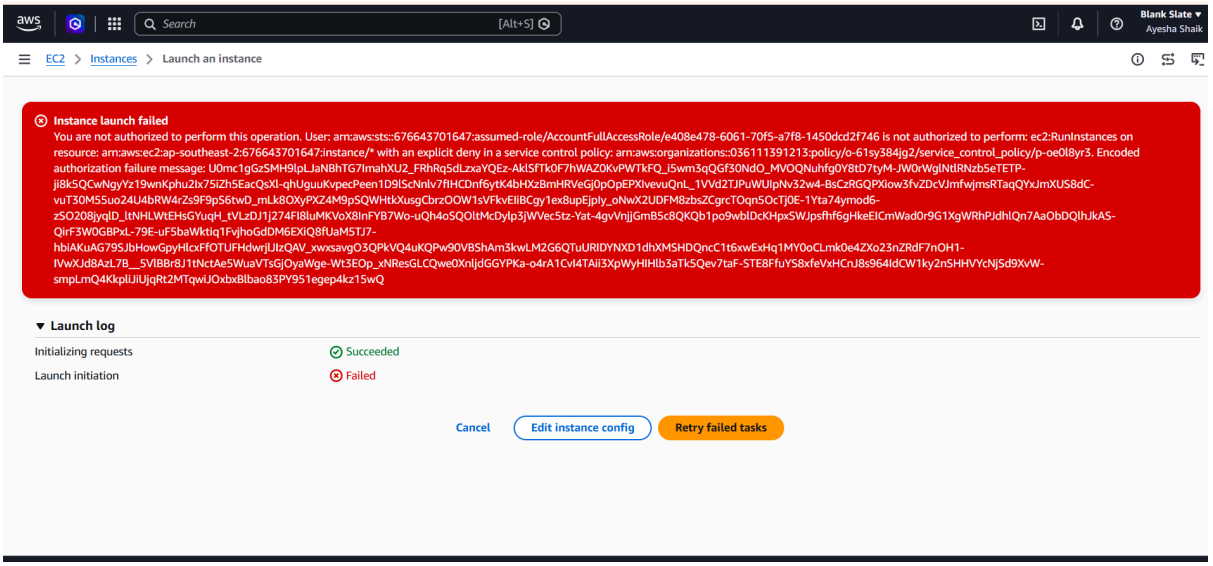


1. I connected to the Ubuntu EC2 instance using the existing .pem key.
2. I generated a new SSH key pair on my local Windows/Git Bash system.
3. I copied the public key to the Ubuntu EC2 instance.
4. I added the public key to the EC2 user's authorized_keys file.
5. I set the required permissions for the .ssh directory and authorized_keys.
6. I exited the EC2 instance.
7. I connected to the EC2 again using the new SSH key.

8. I verified that I could connect without entering a password or specifying the original .pem key.
9. I confirmed that passwordless SSH authentication was working.

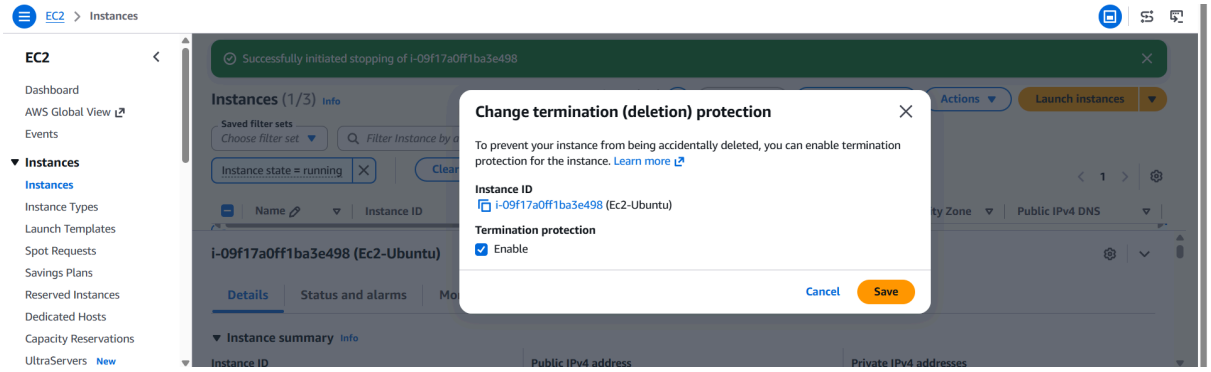
6.Launch any EC2 using the spot purchasing option

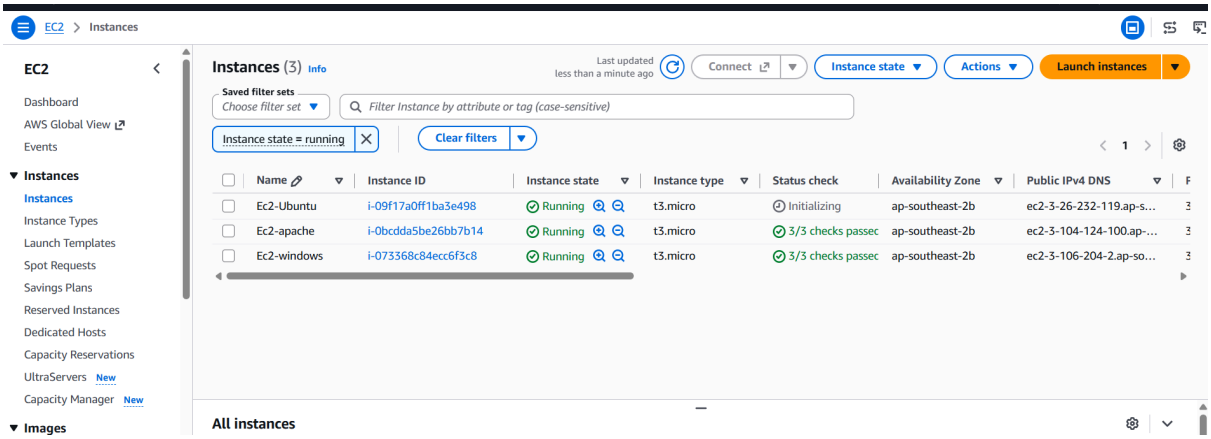




Spot EC2 Launch: Attempted to launch an EC2 instance using Spot purchasing. The request was blocked by an AWS Organization-level SCP/permission restriction. Since this restriction is controlled by the organization administrator, the Spot Instance could not be launched from the current account.

7. Enable termination policy on the EC2 created in Task 2.

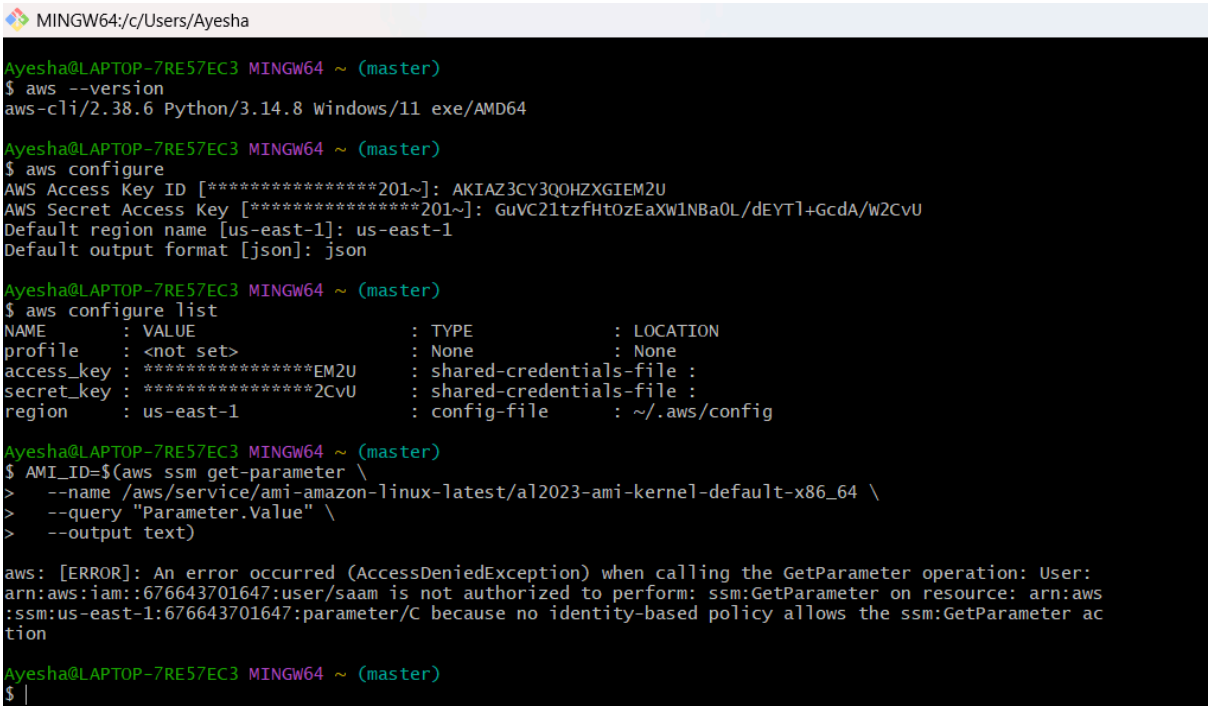




Enable Termination Protection — Steps Only

1. Go to AWS Console → EC2.
2. Click Instances.
3. Select already-created Ubuntu EC2 instance.
4. Click Actions.
5. Select Instance settings.
6. Click Change termination protection.
7. Select Enable.
8. Click Save.

8. Launch one EC2 using AWS CLI



I launched an EC2 instance using the AWS CLI from Windows Git Bash. First, I configured the AWS CLI using the aws configure command by providing my AWS Access Key ID, Secret Access Key, default region as us-east-1, and output format as json. I then verified the configuration using aws configure list. After that, I tried to retrieve the latest Amazon Linux AMI ID using the AWS Systems Manager ssm get-parameter command. However, I received an AccessDeniedException stating that my IAM user was not authorized to perform ssm:GetParameter. The problem occurred because the IAM user did not have permission to read the SSM parameter containing the Amazon Linux AMI information. The AWS CLI configuration and authentication were working correctly; only the required SSM permission was missing.